17. ROS2 Rviz2 Use

1. Introduction to Rviz2

During robot development there are all kinds of functions. If we only analyze the data from the data level, it is hard to quickly understand the effect of the data. For example, for a robot model, we need to know what the model we designed looks like, and where the many coordinate frames inside the model are during the motion.

Another example is robotic arm motion planning and autonomous navigation of a mobile robot: we hope to see the environment around the robot and the planned path, and of course the sensor information, such as cameras, 3D cameras, LiDAR, and so on. The data is used for computation, while the visualized result is what people look at.

Therefore, data visualization can greatly improve development efficiency. Rviz2 is exactly such a data visualization software for robot development. The robot model, sensor information, environment information, and so on can all be handled here.

2. Preparation

First, load the ROS2 Jazzy environment in the terminal that starts the simulation:

```
source /opt/ros/jazzy/setup.bash
```

- If you have a physical robot, you can start rviz on the robot's main control board to practise the content of this section. If you do not have a physical robot, you can start the turtlebot3 simulated robot through the gazebo simulation to simulate topics such as the LiDAR and the camera, which makes the following data visualization convenient.

Note: The installation steps below are not mandatory. If you have a physical robot at hand, you can use the LiDAR information of the real machine directly after setting up the multi-machine communication, and you can freely choose to use the real LiDAR or the virtual simulated robot. The following content is suitable for users without a real robot.

- This section uses the simulated robot as an example to teach the visualization functions of rviz2. Whether it is a real robot or a simulated robot, the rviz2 operation flow is the same.
- Install the TurtleBot3 simulator function package

```
sudo apt install ros-${ROS_DISTRO}-turtlebot3*
```

- Install the ros and gazebo bridging tool

```
sudo apt install ros-${ROS_DISTRO}-ros-gz
```

- Set the environment variable of the turtlebot3 robot type

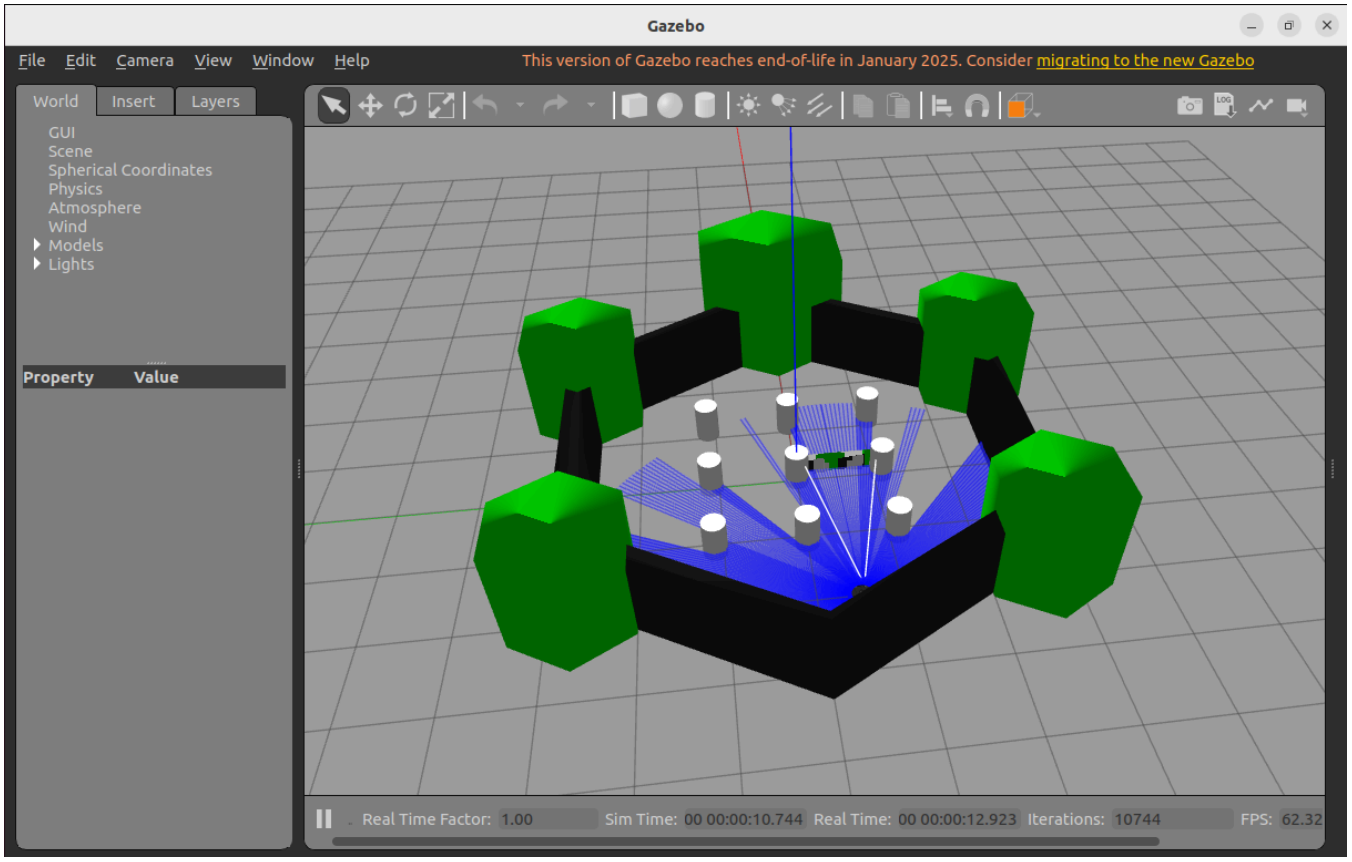
```
export TURTLEBOT3_MODEL=waffle
```

`TURTLEBOT3_MODEL` must be effective in the same terminal in which the simulation is started. You can first run `echo $TURTLEBOT3_MODEL` to confirm that it returns `waffle`, otherwise the launch file will report `'TURTLEBOT3_MODEL'`.

- Start the gazebo simulation environment

```
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```

It is recommended to run it in the VNC graphical desktop. An SSH or RDP terminal without a correct `DISPLAY` may fail to show the Gazebo GUI; on devices with little memory you should also close the other Gazebo processes first to avoid EGL errors or insufficient memory. Do not start TurtleBot3 and the other Gazebo worlds of this tutorial at the same time.

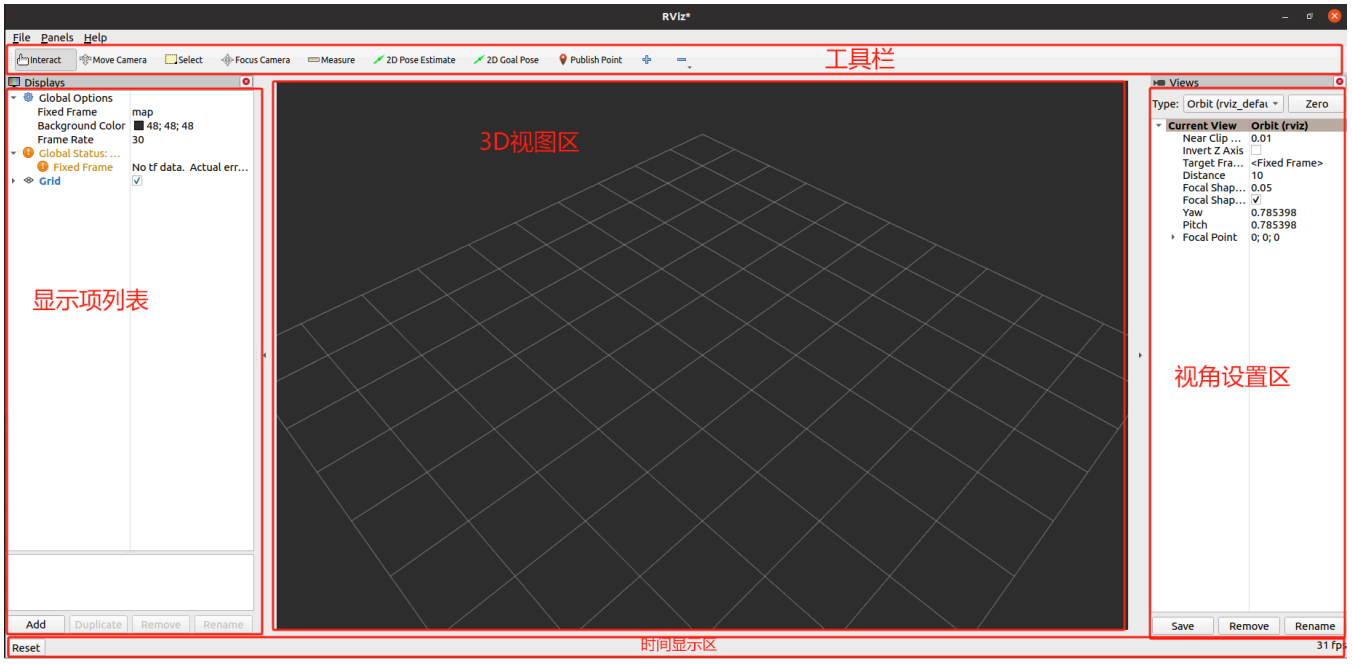


3. Starting Rviz2

Open a terminal, and you can start it with the following command:

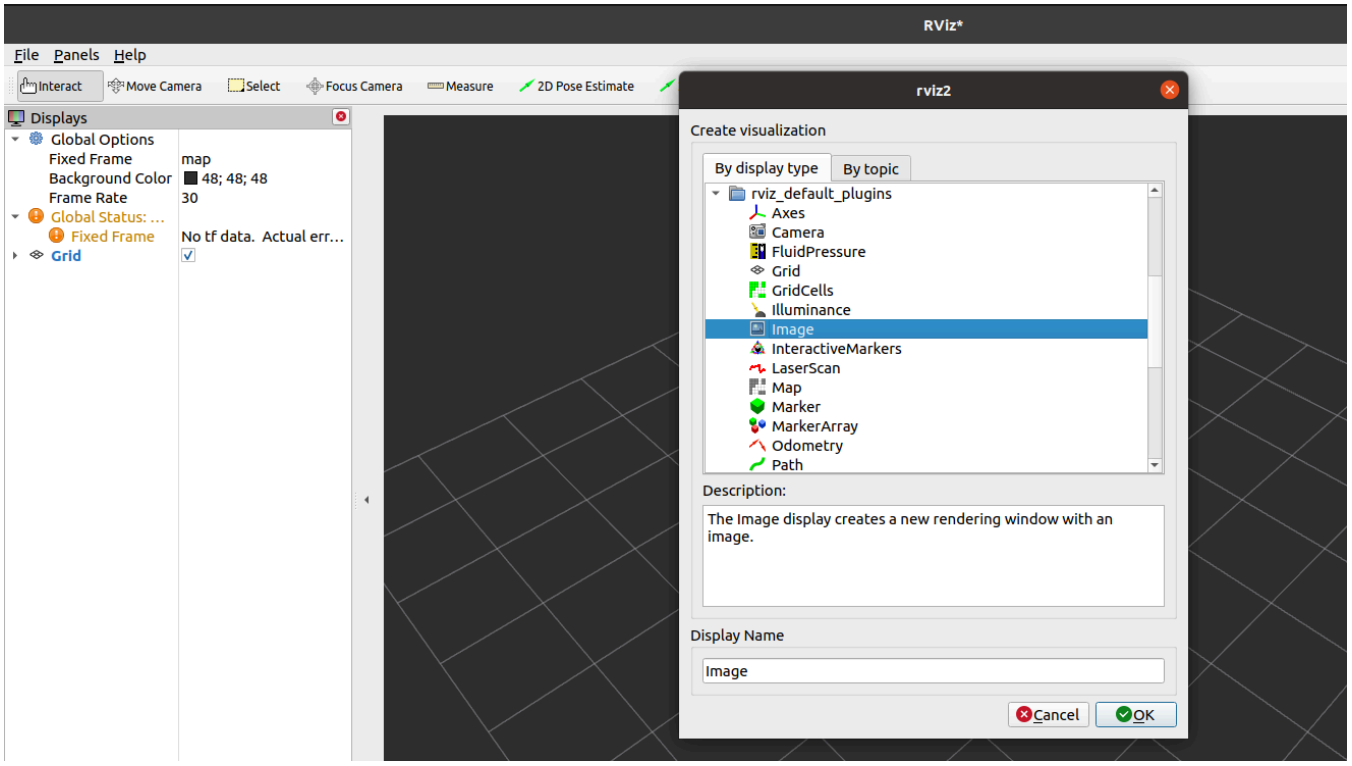
```
rviz2
```

If you start it inside docker, please make sure that the GUI display has been enabled.

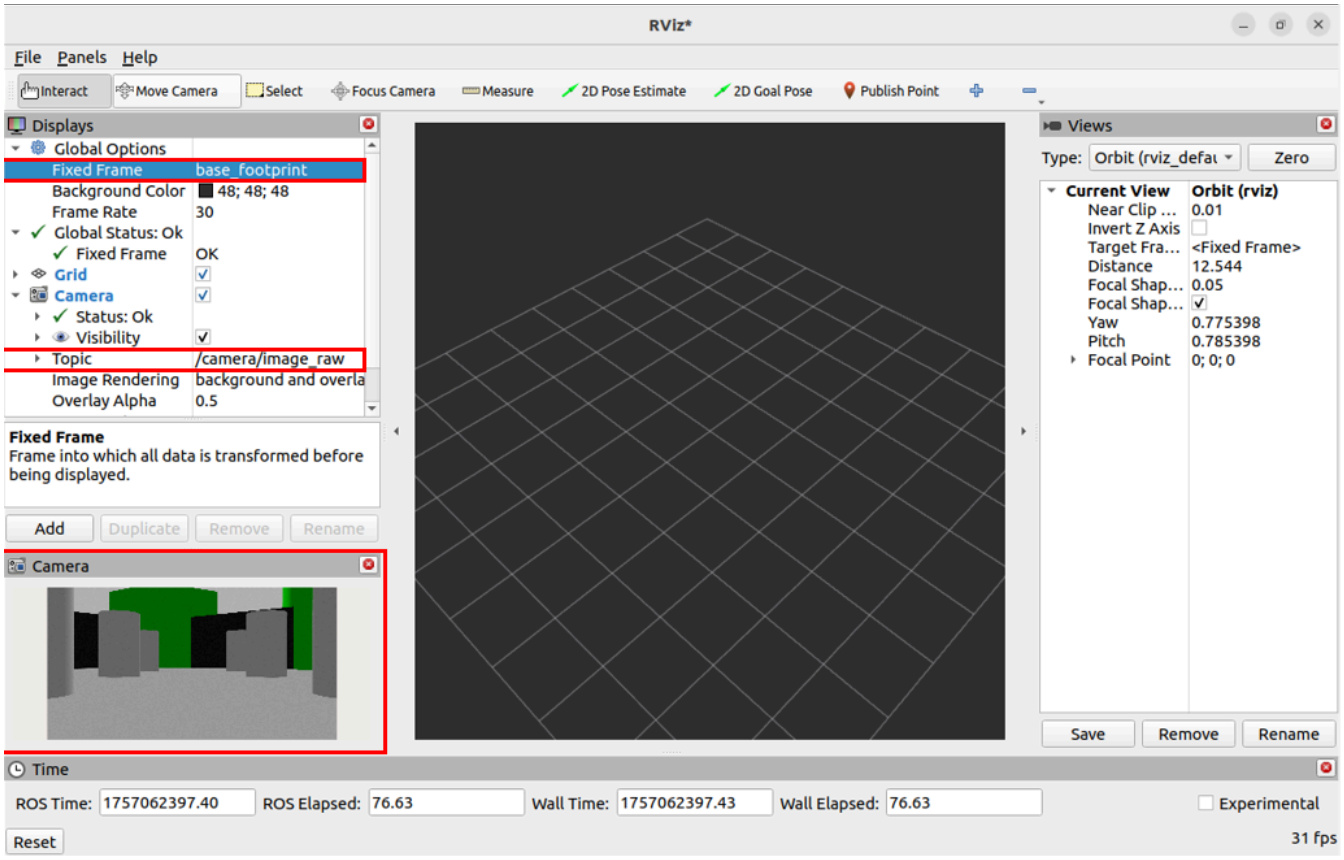


4. Image Data Visualization

Click "Add" in the Displays window on the left, find the Image display item, and after confirming with OK it is added to the display list. Then configure the image topic that this display item subscribes to, and you can smoothly see the camera image of the robot.

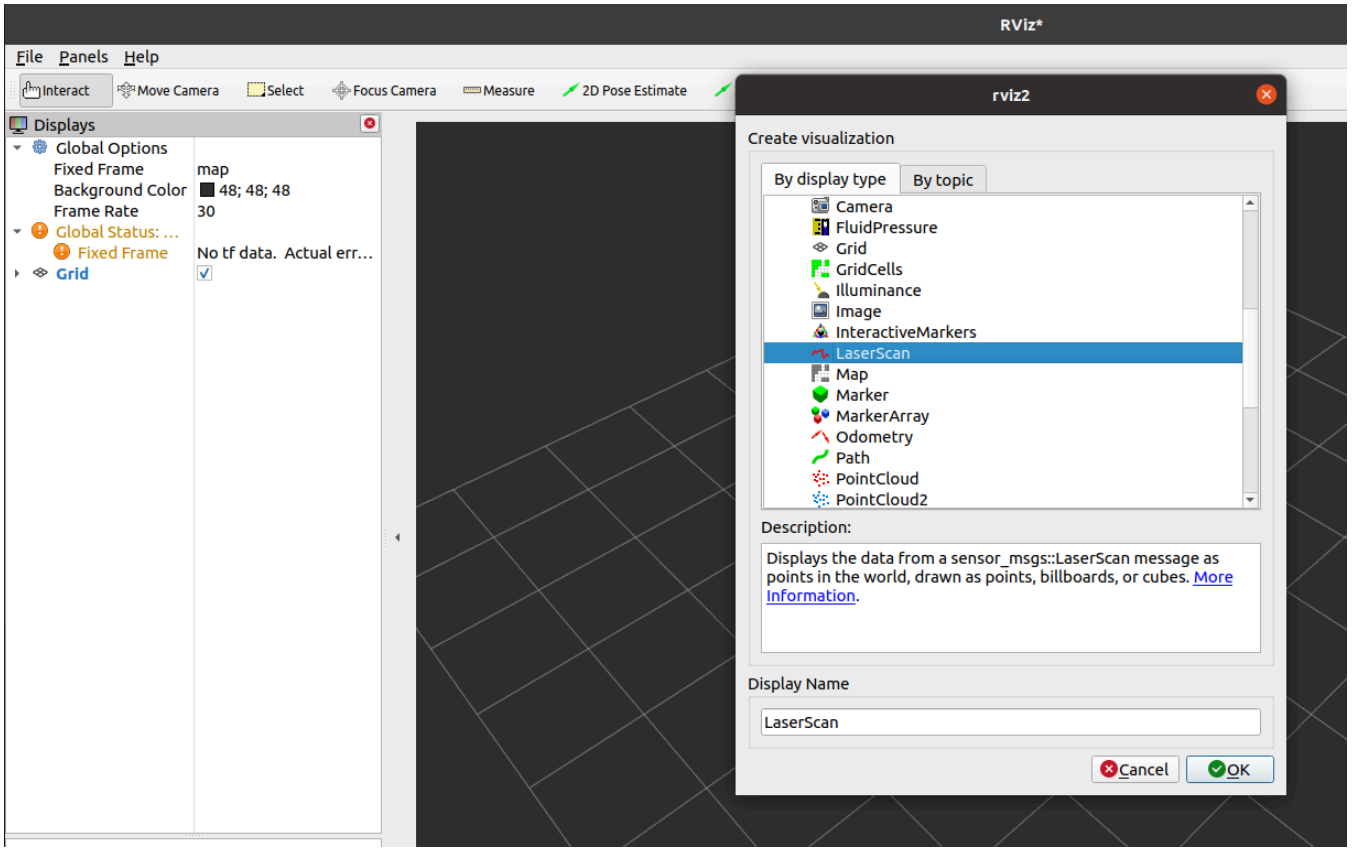


- Set `Fixed Frame` to the `base_footprint` coordinate frame to prevent coordinate transformation errors
- Select the camera colour image topic `/camera/image_raw`
- At this point you can see the view of the current simulated robot in the Camera window



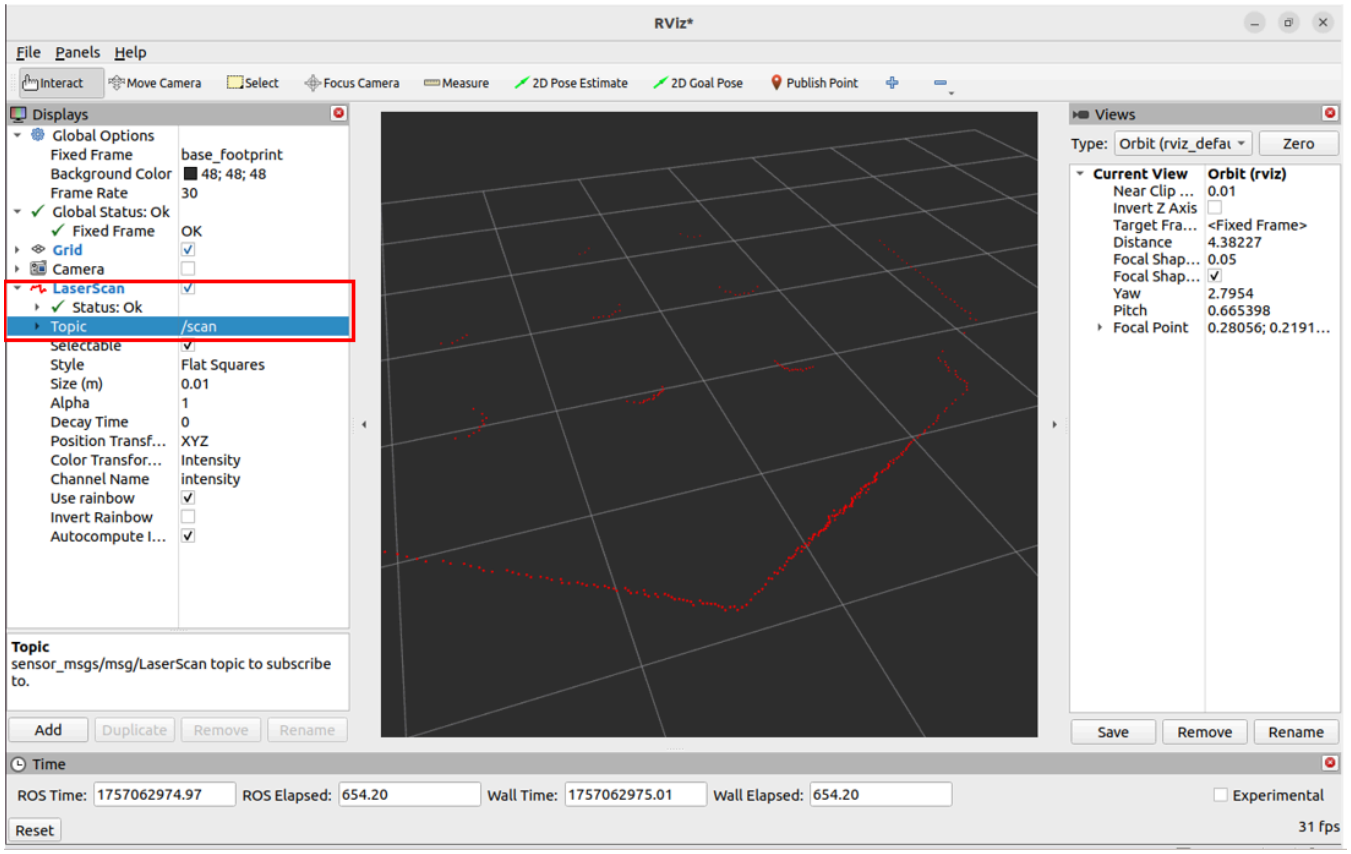
5. LiDAR Data Visualization

Click "Add" in the Displays window on the left, select Laserscan, and then configure the name of the subscribed topic. At this point you can see the laser points.



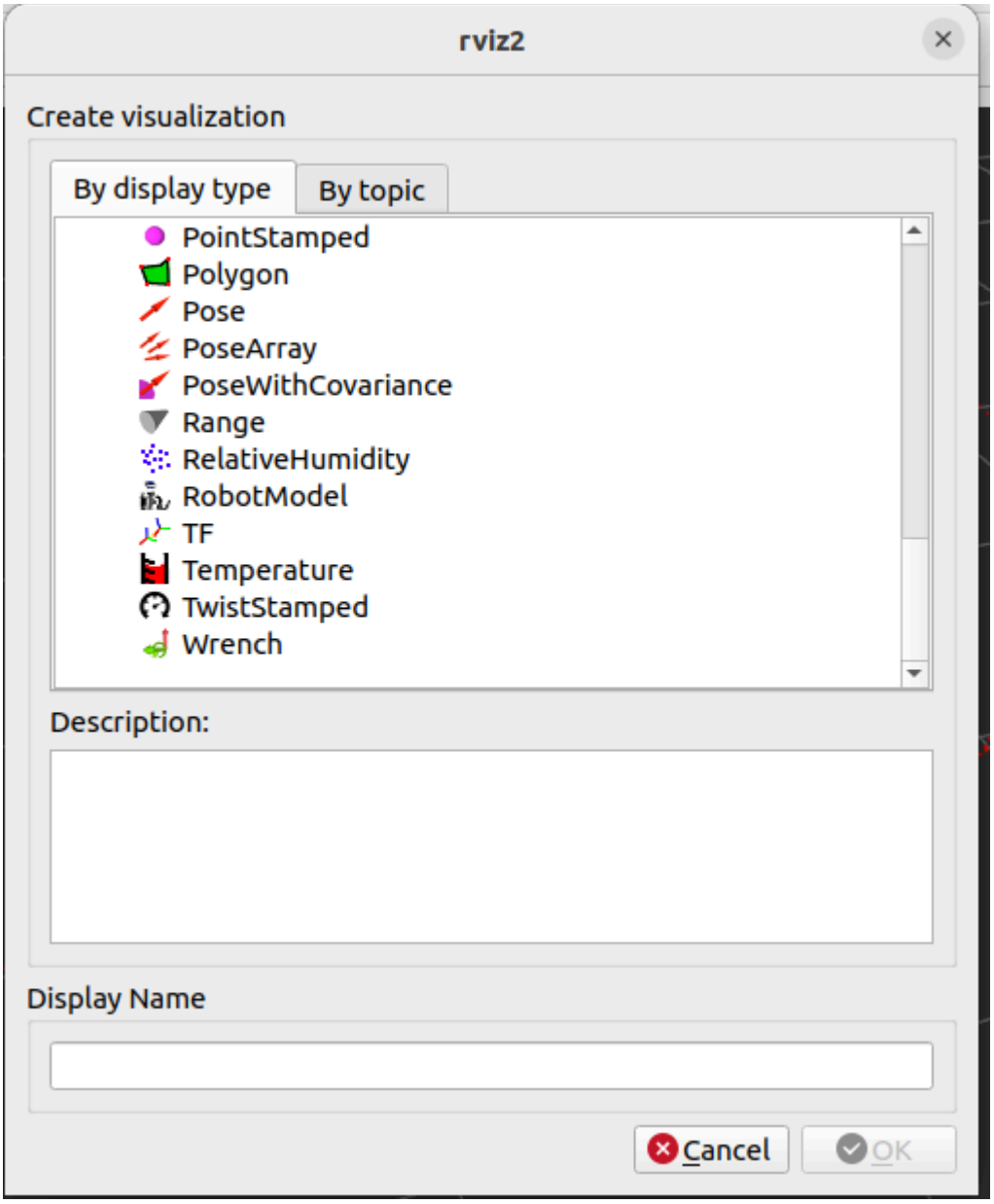
- Set the `Topic` of `LaserScan` to `/scan`

- At this point you can see the point cloud outline of the LiDAR

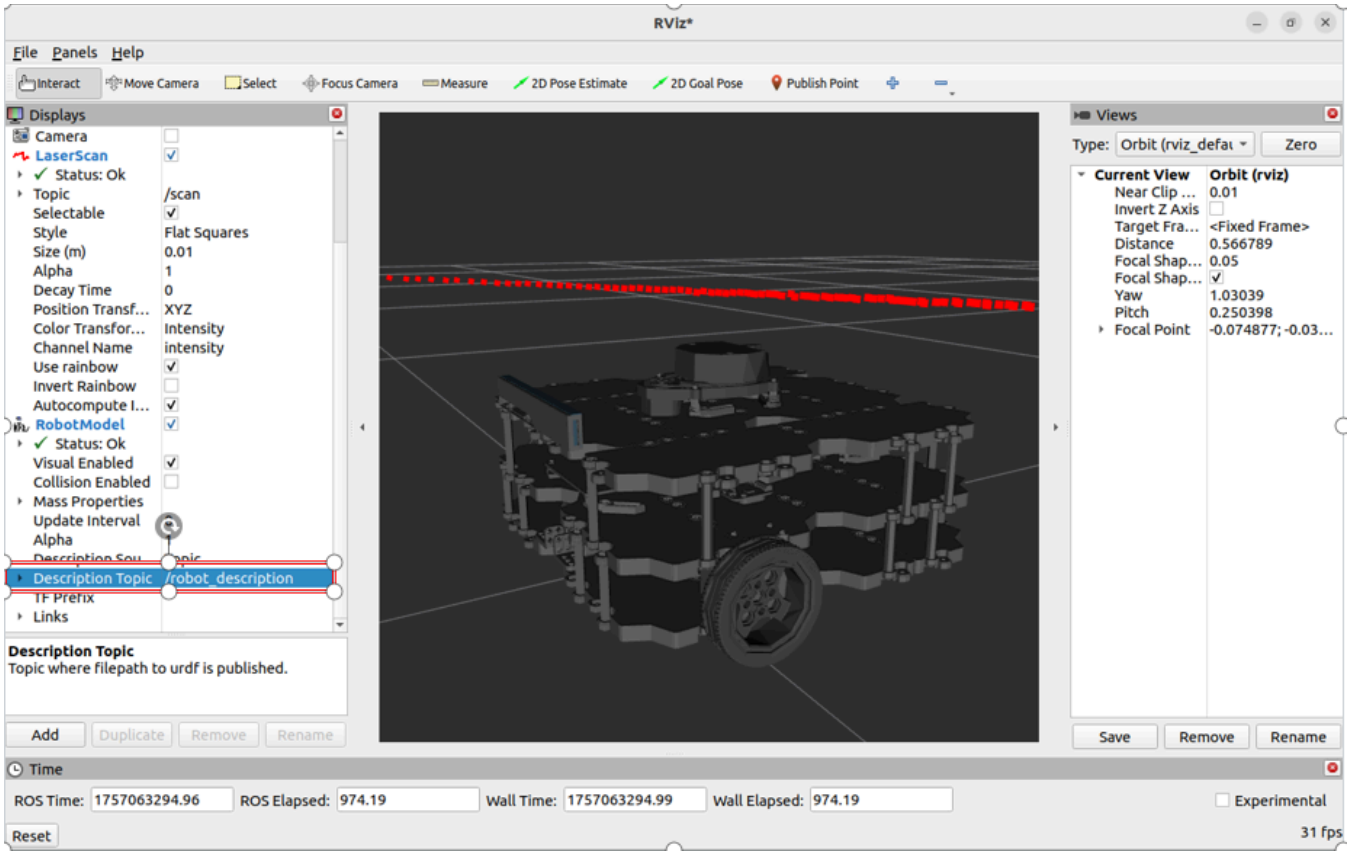


6. Robot Model Visualization

Click "Add" in the Displays window on the left and select RobotModel

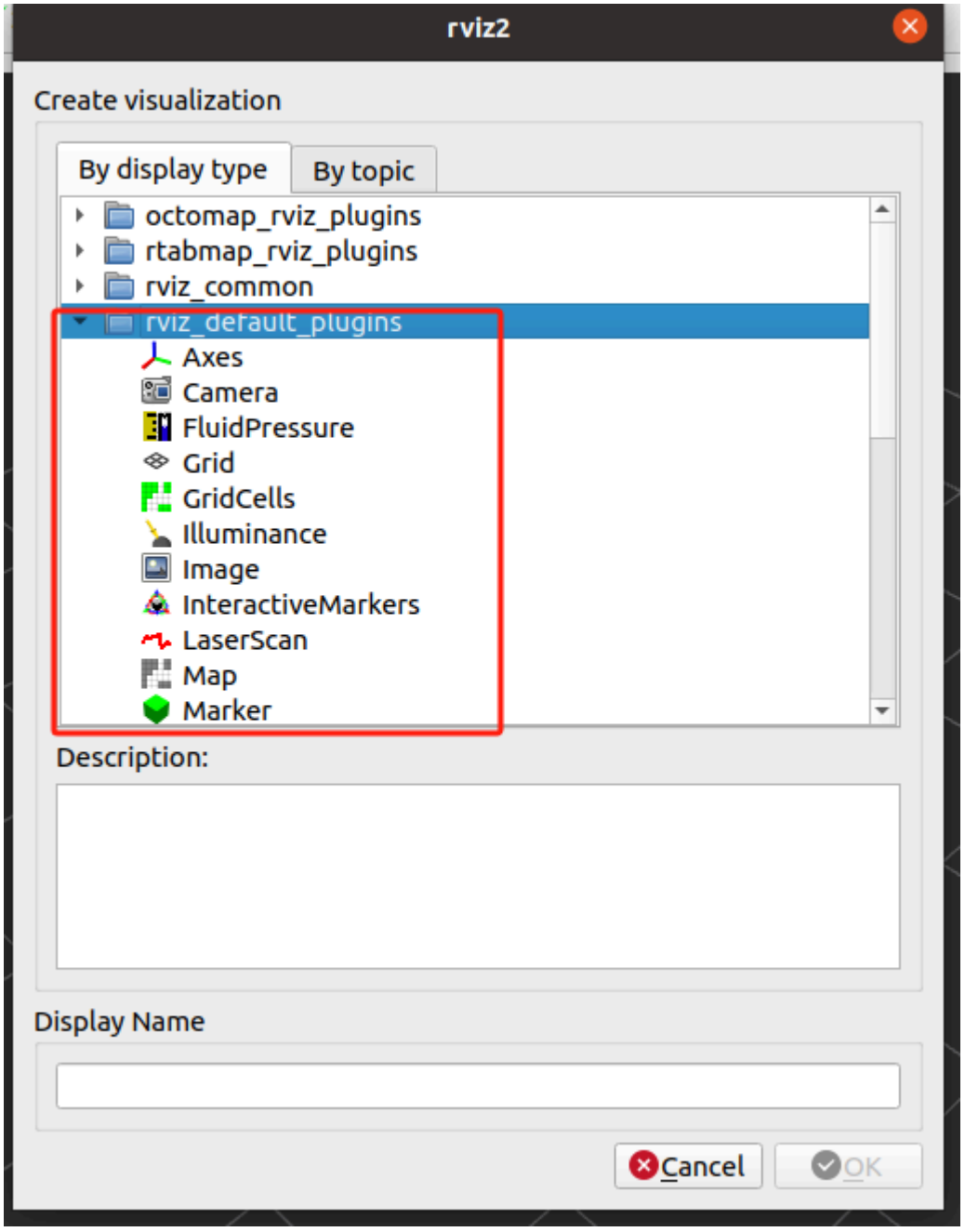


- In the `DescriptionTopic` option of the robot, select the topic `/robot_description`
- At this point you can see the visualization model of the robot in Rviz2



7. Other Data Visualization

rviz_default_plugins lists many commonly used data visualization plugins. You can try them one by one.


rviz2
Create visualization
By display type
By topic
octomap_rviz_plugins
rtabmap_rviz_plugins
rviz_common
rviz_default_plugins
Axes
Camera
FluidPressure
Grid
GridCells
Illuminance
Image
InteractiveMarkers
LaserScan
Map
Marker
Description:
Display Name
Cancel
OK